Alagille Syndrome, Part Seven: The Pruritus Ladder. Measuring Itch, and the Evidence Behind Ursodeoxycholic Acid, Cholestyramine, Rifampicin, Naltrexone and Sertraline.

Imagine a girl of three with Alagille syndrome, confirmed by a disease-causing variant in the gene JAG1. She still has her own liver. Her total bilirubin is about six milligrams per decilitre and her serum bile acids are around two hundred and eighty micromoles per litre, many times the upper limit of normal. For the past year she has taken ursodeoxycholic acid, a bile acid given to make bile flow more freely, and at night she gets hydroxyzine, a sedating antihistamine. She scratches until her skin bleeds, and she sleeps about four hours a night. Her mother keeps an itch diary and rates the itch three and a half on a scale from zero to four on most days.

The team has five questions. Is her mother's score a number worth treating to, and how much of a change would actually matter? Has anything on the treatment ladder really been tried yet? Which of the classic drugs, rifampicin, cholestyramine, naltrexone or sertraline, should come next, at what dose, and for how long before deciding it has failed? A drug of a newer class, an inhibitor of the ileal bile acid transporter, is approved for a child of her age, so should that be the last rung or an early one? And the family is already asking about surgery to divert bile, or about a liver transplant to stop the itch.

The main idea of this chapter is that itch is the symptom that most often drives a family with Alagille syndrome to ask for a transplant, yet the traditional ladder of treatments rests on small and mostly uncontrolled studies. Rifampicin has the most data in children. Naltrexone and sertraline have small randomised trials in adults and small series in children. Ursodeoxycholic acid and antihistamines have almost no evidence as treatments for itch. In practice that means measuring the itch properly, giving each rung a fixed and scored trial, watching the liver while doing it, and remembering that both time and placebo make itch scores fall too. It also means recognising that, since 2021, the only randomised evidence for itch in Alagille syndrome belongs to the newest drug class, which is why the European guideline now places it early rather than last.

Why there is no single drug for cholestatic itch

The itch of cholestasis does not have a single cause. Several candidate mediators have been identified, and each of the classic drugs acts on only one of them.

The first candidate is the bile acids themselves. When bile cannot leave the liver normally, bile acids build up in the blood and in the skin. In 2019, James Meixiong and colleagues, writing in the journal eLife, showed that a receptor called MRGPRX4, found on human sensory nerve fibres that carry itch, is activated by bile acids. The same group showed in a second paper that bilirubin can activate this receptor too. This is the logic behind every treatment that lowers the bile acid pool: resins that bind bile acids in the gut, drugs that block their reabsorption at the end of the small intestine, and operations that divert bile out of the body.

The second candidate is an enzyme called autotaxin. Autotaxin produces a lipid called lysophosphatidic acid, which strongly activates sensory neurons. In 2010, Andreas Kremer and colleagues reported in Gastroenterology that serum from itchy cholestatic patients activated

neurons, that the active substance was lysophosphatidic acid, and that injecting it into the skin of mice made them scratch. In adults, autotaxin activity tracks the intensity of itch better than the level of bile acids does.

The third candidate is a nuclear receptor called the pregnane X receptor. This is not a pruritogen itself, but it is the switch that rifampicin turns on. When the receptor is activated, the liver and gut make more of the enzymes that add hydroxyl and sulphate groups to bile acids, which makes them less toxic and easier to excrete. In 2012, the same Amsterdam group showed in Hepatology that rifampicin also lowers the production of autotaxin through this receptor. This explains why rifampicin can relieve itch even when the measured bile acid level hardly changes.

The fourth candidate is the body's own opioid system. Cholestasis raises the tone of the central opioid pathways, and opioids cause itch. Naltrexone, an oral opioid antagonist, blocks this. The cost is that blocking opioid receptors in a person whose system has adapted to high opioid tone can cause a reaction that resembles opioid withdrawal.

The fifth candidate is serotonin. In a small adult study, sertraline, a selective serotonin reuptake inhibitor, relieved cholestatic itch independently of any effect on mood, which points to a role for central serotonin pathways in how itch is perceived.

The practical consequence is simple. Each old drug hits one pathway, which is why each helps some children partly and few children completely. Combining drugs that act on different pathways is rational. Replacing one weak drug with another that works the same way is not. And the agents that lower the bile acid pool most directly are the ones with the strongest controlled evidence.

The itch is not in the bilirubin

A small study from Paris and Amsterdam, published by Kremer, Emmanuel Gonzales and colleagues in the Journal of Pediatric Gastroenterology and Nutrition in 2016, looked at this question in children. They studied forty-five children. Fourteen had cholestasis with itch, and ten of those fourteen had Alagille syndrome. Nine had a defect of bile acid synthesis, a group of conditions in which children are jaundiced but cannot make normal bile acids. Twenty-two were healthy controls.

Average autotaxin activity was about sixteen units in the itchy children, about ten in the children with bile acid synthesis defects, and between seven and eight in the healthy children. In other words, the itchy children had roughly twice the autotaxin activity of healthy children, while the jaundiced but non-itchy children sat in between. Itch intensity correlated moderately with autotaxin, with a correlation coefficient of about zero point six, and in these children it correlated even more closely with total serum bile salts, with a coefficient of about zero point eight. It did not correlate with bilirubin. It is worth knowing that all the itchy children were already taking rifampicin and ursodeoxycholic acid, so their autotaxin levels were measured on treatment.

The children with bile acid synthesis defects make an elegant natural control. They are cholestatic and jaundiced, yet they do not make the usual toxic bile acids, and they do not itch. The lesson for the clinic is that itch, and the response to treatment, should never be judged by the bilirubin.

Measuring itch

A treatment ladder only works if each rung is judged, and that requires a score. Several instruments are used in pediatric cholestasis.

The most important in Alagille syndrome is called ItchRO, which stands for Itch Reported Outcome. It was developed by Binita Kamath, Benjamin Shneider and colleagues and published in the journal The Patient in 2018. It has two versions. The observer version is completed by a caregiver for younger children, and the patient version is completed by children from about nine years of age. Each version is a single question scored from zero to four, running from not itchy at all to extremely itchy. It is completed twice a day, in the morning for the night before and in the evening for the day, because families found a twelve-hour recall easier than a full day. The daily score is the worse of the two. In analyses of maralixibat, a fall of one point or more on the caregiver version has been used to define a responder.

Other instruments include the clinician scratch scale, a zero to four score that a doctor assigns from the skin at a clinic visit. The authors of the North American natural history study noted that it correlates poorly with ItchRO. There is also a caregiver scratching score, again from zero to four, that served as the main endpoint of the odevixibat trials. Finally, families judge treatment largely by sleep: how often the child wakes, whether the child ends up in the parents' bed, and quality of life scores.

The practical point is to ask the caregiver to keep the twice-daily diary for two weeks before a new drug and throughout its trial, and to record sleep and the state of the skin. A rung without a baseline score is a rung that can never be judged, and that is how children end up on a drug for years that never worked.

Itch improves with time, and on placebo

Before giving credit to any new drug, it is important to know how much itch changes without one. Two kinds of data answer that question.

The first is natural history. The Childhood Liver Disease Research Network, a North American consortium, followed fifty-nine children with Alagille syndrome and itch, and Benjamin Shneider and colleagues reported the results in Hepatology Communications in 2022. On the clinician scratch scale, which runs from zero to four, the average score fell by about one point after one year and by about one and a half points after two years, while the use of the usual itch medicines stayed stable. By the second year, seventy-three percent had a clinically relevant fall, and nine of forty children had no itch at all.

The second is the placebo arm of randomised trials. In the ASSERT trial of odevixibat in Alagille syndrome, the scratching score fell by about one point seven with the drug and by about zero point eight with placebo, a significant difference. In adults with primary biliary cholangitis, the GLISTEN trial of linerixibat, published in 2026, found that on a ten-point itch scale the score fell by about two point nine with the drug and by about two point two with placebo, so the drug added roughly three quarters of a point.

The scales differ, so the numbers should only be compared within each study. But the message is consistent. Itch improves with time and with the attention of a trial. Every open-label series of rifampicin, naltrexone or sertraline therefore overstates the drug, and before crediting the newest rung we should ask whether time or the trial itself did the work.

The real-world ladder before the newer drugs

What did the full traditional ladder achieve? The best answer in Alagille syndrome comes from King's College Hospital in London, published by Vandana Kronsten, Emer Fitzpatrick and Alastair Baker in the Journal of Pediatric Gastroenterology and Nutrition in 2013. They reviewed sixty-two children with Alagille syndrome diagnosed between 1995 and 2010.

Fifty-one of the sixty-two, or about eighty-two percent, had itch, and fifty received medicines for it. Most received several. Forty received ursodeoxycholic acid and thirty-nine received rifampicin. Eighteen received cholestyramine, fourteen received naltrexone, and thirteen received alimemazine, a sedating antihistamine. Smaller numbers received non-sedating antihistamines, ondansetron, and phenobarbitone.

The outcomes are sobering. Medicines alone controlled the itch in twenty-one of the fifty-one itchy children, about forty-one percent. The itch resolved permanently in twenty children, about thirty-nine percent, but eleven of those twenty had needed a liver transplant to get there. In ten children, about one in five, the itch was still significantly harming quality of life. Sixteen of the sixty-two were listed for transplantation. The honest summary is that, before the newer drugs, the full ladder controlled itch with medicines in only about two children out of five.

The lower rungs: skin care, ursodeoxycholic acid and antihistamines

The first steps are worth doing but should not be mistaken for treatment of severe itch. Skin care means short nails, cotton sleep-suits, a cool bedroom, emollients, and treatment of broken and infected skin. It is cheap, safe and largely unstudied.

Ursodeoxycholic acid is usually given at fifteen to twenty milligrams per kilogram per day. It promotes bile flow and is almost universally used in cholestatic children, but there are no controlled data showing that it relieves itch in Alagille syndrome. The 2024 European guideline on genetic cholestatic liver diseases lists it as an off-label treatment with low efficacy for itch, and adult reviews note that it can occasionally make itch worse.

Antihistamines such as hydroxyzine or alimemazine are used early by almost everyone, but cholestatic itch is not driven by histamine. Their benefit is sedation, which can buy sleep, rather than relief of itch. The European guideline rates them as marginal. Phenobarbital, an older enzyme inducer, causes sedation and behavioural effects and has largely been replaced by rifampicin.

The practical point is that if a child with an itch score of three or more is taking only ursodeoxycholic acid and a night-time antihistamine, the active treatment ladder has not yet started. Keep the antihistamine only if it clearly helps sleep, and do not count months spent on it as a failed step.

Cholestyramine

Cholestyramine is an anion exchange resin. In the intestine it binds conjugated bile acids so that fewer of them are reabsorbed and returned to the liver. It is the oldest form of interrupting the circulation of bile acids between the gut and the liver. In adult guidance, including the 2009 European guideline on cholestatic liver disease and the 2018 American guidance on primary biliary cholangitis, it is listed first, mainly on the basis of long use and small old studies. The 2024 European guideline on genetic cholestasis rates its efficacy as marginal.

It binds more than bile acids. It also binds ursodeoxycholic acid, the fat-soluble vitamins and many other drugs, so those must be given at least an hour before or four hours after the resin. Its taste and texture are poor, and it causes bloating and constipation. Because it exchanges chloride for bile acid, it can cause a hyperchloraemic metabolic acidosis in small children, meaning a fall in bicarbonate with a rise in chloride. In a child with Alagille syndrome who is already short of vitamin D and whose kidneys may already be losing bicarbonate, cholestyramine can make both problems worse.

A reasonable approach is a short, scored trial in an older child who can actually take it, with the doses separated from other medicines, and with vitamin D, the international normalised ratio, and bicarbonate rechecked within a few weeks. If the diary has not moved after about a month, stop it.

Rifampicin: the evidence

Rifampicin is the best-studied of the old drugs in children, and three pediatric studies are worth knowing.

In 1990, Harvey Cynamon and colleagues reported in Gastroenterology a six-week double-blind crossover trial against placebo in five children with chronic cholestasis. Rifampicin relieved itch in all five compared with placebo, and the benefit lasted through six months of continued treatment, with no complications attributed to the drug.

In 1993, Gerardine Gregorio, Giorgina Mieli-Vergani and colleagues at King's College Hospital reported in the Archives of Disease in Childhood a retrospective review of thirty-three children treated with a median dose of five milligrams per kilogram per day. Twenty-one of them had Alagille syndrome. Itch resolved completely in about fifteen percent and partly in about thirty-six percent, so roughly half had no useful response. In the children with Alagille syndrome, alkaline phosphatase fell significantly at one and at six months. No side effects were reported.

In 1999, Baruch Yerushalmi, Ronald Sokol and colleagues in Denver reported in the Journal of Pediatric Gastroenterology and Nutrition an open study of twenty-four children with severe itch that had not responded to at least two months of ursodeoxycholic acid, antihistamines or phenobarbital. Six had Alagille syndrome and thirteen had biliary atresia. At ten milligrams per kilogram per day in two doses, ten children had complete relief, twelve partial relief and two none. The pattern mattered: complete relief was much more common in extrahepatic cholestasis, about sixty-four percent, than in intrahepatic cholestasis such as Alagille syndrome, about ten percent, while partial relief was the usual result in intrahepatic disease. No toxicity was seen.

The reading for Alagille syndrome is that rifampicin usually gives partial relief, perhaps one or two points on the diary, and rarely a cure.

Rifampicin: giving it safely

The pediatric series used five to ten milligrams per kilogram per day, usually in two doses, while adult guidance uses one hundred and fifty to three hundred milligrams twice a day. A practical approach is to start at five milligrams per kilogram per day and increase to ten after about two weeks, without exceeding the adult maximum of six hundred milligrams a day.

The main concern is the liver. In 2002, Martin Prince, Alastair Burt and David Jones reported in Gut that among adults with primary biliary cirrhosis treated with rifampicin for itch at one centre,

significant hepatitis occurred in about seven percent, with a confidence interval of roughly three to nineteen percent. Two of the three affected patients had impaired synthetic function, and one needed a liver transplant. The pediatric series reported no hepatotoxicity, but they were small. Liver tests, bilirubin and the international normalised ratio should therefore be checked at about two, four and eight weeks and then every three months, and the drug stopped if there is a clear rise. Rifampicin is also a strong inducer of the drug-metabolising enzyme cytochrome P450 3A4, so every other medicine should be reviewed, and it can speed up the breakdown of vitamin D, so twenty-five hydroxy vitamin D should be checked. It turns urine, tears and sweat orange and can stain contact lenses, and it rarely causes haemolysis or a low platelet count, so a full blood count belongs with the liver tests. Response usually appears within weeks; a fair judgement can be made on the diary at four to six weeks, and the drug should be stopped if the score has fallen by less than a point. Rifampicin is the default second rung in most centres, but it is a drug with a hepatitis signal given to children whose liver is the problem, so every course needs a written monitoring plan from the first day.

Naltrexone

Naltrexone has the best controlled adult evidence of the older add-on drugs. In 1997, Frank Wolfhagen and colleagues in the Netherlands reported in Gastroenterology a double-blind trial in sixteen adults with chronic cholestatic itch. Naltrexone fifty milligrams a day for four weeks reduced daytime itch by about fifty-four percent compared with eight percent on placebo, and night-time itch by about forty-four percent compared with seven percent. Four naltrexone-treated patients had symptoms resembling opioid withdrawal, three of them transient.
In children the evidence is thinner. The largest series, from the Children's Hospital of Philadelphia and published in the Journal of Pediatric Pharmacology and Therapeutics in 2023, reviewed thirty-nine inpatients treated between 2008 and 2021, eight of whom had Alagille syndrome. The median dose was about one point nine milligrams per kilogram per day, with a very wide range, usually given once a day. No itch score was used, so efficacy could not be measured, and naltrexone was usually added as a fourth drug on top of three others rather than replacing anything. Two practical points follow. Because of the withdrawal-like reaction, start at about a quarter of the target dose and increase over one to two weeks. And because naltrexone blocks opioid pain relief, it must be stopped before planned surgery or painful procedures, and the anaesthetist must be told. The honest position is that naltrexone is a reasonable add-on with a scored trial, and it should be stopped if the diary has not moved.

Sertraline

In adults, Marlyn Mayo and colleagues reported in Hepatology in 2007 a study of twenty-one patients with itch from chronic liver disease. After an open dose-finding phase, twelve were randomised. At seventy-five to one hundred milligrams a day, sertraline improved itch while placebo did not, and the difference was significant.
In children, Alice Thébaut, Emmanuel Gonzales and colleagues in France reported in the Journal of Pediatric Gastroenterology and Nutrition in 2017 a series of twenty children with refractory cholestatic itch, most already on rifampicin and other drugs. At a median dose of about two point

two milligrams per kilogram per day, itch improved after three months in fourteen of the twenty. The median itch score on a ten-point scale fell from eight to five. In nine of those fourteen, scratch marks and sleep also improved. Six children had non-severe side effects, and three stopped the drug.

A three-point fall on a ten-point scale is meaningful, but this was an open series without a control group, and time and the expectation of benefit account for part of it. Sertraline belongs as an add-on for refractory itch rather than a first rung. In adolescents it carries the boxed warning about suicidal thinking in young people taking antidepressants, so mood should be discussed and monitored, and its serotonergic interactions remembered.

The ladder at a glance

Seen side by side, the rungs look like this. Ursodeoxycholic acid acts by promoting bile flow, at fifteen to twenty milligrams per kilogram per day, with no controlled evidence for itch and little harm. Cholestyramine binds bile acids in the gut, with only old small studies behind it, and its price is taste, acidosis and binding of vitamins and drugs. Rifampicin acts through the pregnane X receptor and autotaxin, at five to ten milligrams per kilogram per day, with a small pediatric crossover trial and two series of thirty-three and twenty-four children, and its price is a hepatitis risk of around seven percent in adults. Naltrexone blocks opioid tone, titrated to a median of about two milligrams per kilogram per day in the Philadelphia series, backed by an adult randomised trial of sixteen patients, at the price of a withdrawal-like reaction and blocked pain relief. Sertraline acts on serotonin, at roughly one to four milligrams per kilogram per day, with a pediatric series of twenty and an adult randomised phase of twelve, at the price of activation and the boxed warning. Finally, the inhibitors of the ileal bile acid transporter block reabsorption of bile acids at the end of the small intestine, backed by randomised data in Alagille syndrome, with diarrhoea and fat-soluble vitamin loss as their main costs. The rule is to combine rungs with different targets, give each a four-to-six-week trial on a diary, and remove what did not work.

The top rungs, and what changed after 2021

The only randomised evidence for itch in Alagille syndrome comes from the newest rung. In the ICONIC study of maralixibat, published by Emmanuel Gonzales and colleagues in The Lancet in 2021, thirty-one children took the drug for eighteen weeks, after which some were randomly switched to placebo for four weeks. Those switched to placebo saw their itch worsen by about one point seven points and their bile acids rise by about ninety-four micromoles per litre, while those who stayed on the drug did not. By week forty-eight, itch had improved by about one point six points from baseline.

In the ASSERT trial of odevixibat, published by Nadia Ovchinsky and colleagues in The Lancet Gastroenterology and Hepatology in 2024, fifty-two children were randomised two to one to the drug or placebo. As already described, the scratching score fell by about one point seven with odevixibat and by about zero point eight with placebo after twenty-four weeks, and diarrhoea occurred in about twenty-nine percent on the drug compared with six percent on placebo.

Surgical diversion of bile is far less attractive in Alagille syndrome than in some other genetic cholestatic diseases. In the Global Alagille Alliance study, presented as an abstract in 2023,

sixty-two of one thousand six hundred and seventy-three patients, under four percent, had undergone surgical biliary diversion. All of them had itch before surgery and about ninety-two percent still had it afterwards, and the operation was not followed by better survival with the native liver. Liver transplantation does cure the itch, but it is a transplant performed for a symptom, and that decision belongs to a later lesson.

The 2024 European guideline on genetic cholestatic liver diseases now recommends that, when available, an inhibitor of the ileal bile acid transporter should be offered to patients with Alagille syndrome and cholestatic itch, a strong recommendation based on level two evidence. The details of these trials and of dosing are the subject of the next lesson. Here the point is about position: the newer drug is no longer the last rung.

What the guidelines say

The guidelines agree on some points and differ on others. For the order of the old rungs, the 2009 European guideline for cholestatic liver disease and the American guidance on primary biliary cholangitis describe the same adult sequence: a bile acid binding resin, then rifampicin, then naltrexone, then sertraline. The 2024 European guideline for genetic cholestasis gives no order, but rates rifampicin as moderately effective and the others as marginal. The 2017 North American and European guideline on cholestatic jaundice in infants does not address itch treatment. On the order, the documents are aligned.

On rifampicin safety, the adult guidance warns about hepatotoxicity and advises monitoring and caution when bilirubin is high, and the European guideline notes its off-label use and its action through the pregnane X receptor. Here too they are aligned.

On antihistamines they differ. American guidance allows them, while European guidance does not recommend them except perhaps for sedation, and the 2024 European guideline rates them as marginal. On ursodeoxycholic acid as a treatment for itch they also differ in emphasis, from not an antipruritic at all in the European adult guideline, to limited effect in American guidance, to low efficacy and off-label use in the 2024 European pediatric document.

The material difference is the place of the newer drugs. They did not exist when the older guidelines were written. In adults, the joint guideline from the American Gastroenterological Association and the American Association for the Study of Liver Diseases, published in October 2026, suggests considering linerixibat for moderate to severe itch in primary biliary cholangitis. In children with Alagille syndrome, the 2024 European guideline recommends offering an inhibitor of the ileal bile acid transporter when available. Finally, there is a gap: no guideline gives a pediatric order, dose, itch scale or trial length for the old drugs.

Building a practical ladder

Putting the evidence together gives a practical sequence, though the order of the middle steps is expert opinion rather than graded evidence. Step one is to measure: a two-week ItchRO diary with notes on sleep and skin, together with vitamins, the international normalised ratio and bicarbonate, and a check that ursodeoxycholic acid and skin care are actually being used. Step two is to lower the bile acid load: an ileal bile acid transporter inhibitor where it is available and the child qualifies, with rifampicin as a bridge or as the alternative where it is not. Step three is to add a

different target: rifampicin with liver monitoring if it has not yet been used, and cholestyramine only if the child tolerates it and the doses are separated. Step four is central treatment: naltrexone titrated, or sertraline, as add-ons for residual itch, each with a scored four-to-six-week trial. Step five is escalation: a persistent score of three or more with lost sleep should be discussed at the transplant meeting, while surgical diversion is rarely the answer in Alagille syndrome.
Rifampicin comes first among the old drugs because it has the most pediatric data. The newer drug comes early because it has the only controlled pediatric data. And a rung that has failed should be removed before the next is added, or the child ends up on five drugs and nobody knows which one works.

Traps and pearls

Several traps recur. Counting a year of ursodeoxycholic acid and a night-time antihistamine as failed medical therapy. Changing drugs without a baseline or follow-up score, so that no rung is ever judged. Crediting the newest drug for an improvement that time or placebo would have given. Using the bilirubin to judge itch. Starting rifampicin without a schedule of liver tests, or forgetting its effects on vitamin D and other drugs. Giving cholestyramine at the same time as ursodeoxycholic acid and vitamins and then wondering why vitamin levels and bicarbonate fall. Starting naltrexone at the full dose and calling the withdrawal-like reaction intolerance, or leaving it in place before surgery. Prescribing sertraline to an adolescent without discussing mood. And keeping the newer drug for the last rung after years of poor sleep.
A few numbers are worth carrying into clinic. In children, itch tracks bile acids and autotaxin, not bilirubin. The ItchRO observer score runs from zero to four, twice a day, and a one-point fall defines a responder. In the North American natural history cohort, the clinician itch score fell by about one point at one year and one and a half points at two years on stable treatment. Before the newer drugs, medicines controlled itch in about two of every five children with Alagille syndrome at King's College Hospital, and about one in five still had impaired quality of life. Rifampicin at five to ten milligrams per kilogram per day gave complete relief in about fifteen percent and partial relief in about a third in the King's series. Rifampicin hepatitis occurs in about seven percent of adults, so monitoring starts at week two. Naltrexone cut daytime itch by about half compared with less than a tenth on placebo in adults, must be titrated, and blocks opioid pain relief. Sertraline helped fourteen of twenty children at a median of about two milligrams per kilogram per day. Stopping maralixibat made itch worse by about one point seven points, and odevixibat beat placebo by about one point seven against zero point eight. And the European guideline now says to offer one of these drugs to children with Alagille syndrome and itch when it is available.

What to do differently in clinic

Return to the girl of three. Her itch has not really been treated yet, because ursodeoxycholic acid and hydroxyzine are background measures. Start a two-week ItchRO diary, record sleep and skin, and check her vitamins, the international normalised ratio and bicarbonate. Offer an ileal bile acid transporter inhibitor as the next step, with rifampicin under a written liver-test plan as the alternative or as an add-on. Give each new drug a scored trial of four to six weeks, remove the ones that do not move the diary, and add naltrexone or sertraline only as targeted add-ons. If, after all of

this, the score stays at three or more and she still cannot sleep, take her case to the transplant meeting with the diaries in hand.

The deeper change is in how we think about the ladder. It is not a list of drugs to work through in order over years. It is a set of trials, each measured, each with a clear stopping rule, built on the understanding that itch has several mediators, that time and placebo help too, and that the strongest evidence now sits with the drugs that lower the bile acid pool.